

EP 2540 — Computations in Engineering Physics

Computing Project Guidelines

Department of Physics and Engineering
John Carroll University
Spring 2026

Outline

Students are required to complete a group project worth 25% of the final grade in the course. This project is an opportunity for students to explore a problem of interest and to use the tools of the class to solve that problem.

Group Structure

Groups will consist of two or three students. Groups will be assigned by the instructor.

Deliverables and Grade Breakdown

To receive a complete grade for the project, each group must complete the following items.

Project Proposal (10%): Groups will submit a project proposal outlining

- The problem they wish to solve,
- How they might formulate the problem mathematically,
- The methods they plan to use,
- What code components they intend to write, and
- Which group member is responsible for what portions of the work.

For the last item, it is important to think about your strengths and weaknesses are, and how you plan to divide up the work to achieve your group goals.

Project Proposal Meeting (5%): After submitting their proposal, groups will meet with the instructor to discuss the feasibility of the project.

After having their meeting, students may resubmit their project proposal for a complete regrade.

Check-In Meeting (5%): Students will have a second meeting with the instructor in the middle of the semester to discuss progress and/or roadblocks on their project.

Python Code (35%): The Python code may consist of a single “.py” file, or several files in a library. The code must run on the instructor’s computer. The code should be annotated with comments so that the instructor may easily follow what each step is doing. Any Python scripts used for generating figures or movies should also be provided.

Group Presentation (40%): During the final exam period, students will present their work to the instructor and the rest of the class. Student presentations must be 15 minutes. Presentations will be graded based on a rubric which will be provided ahead of time.

Individual Reflection (5%): Following the group presentation, each student will write their own reflection statement about how the project went. In particular, address the following:

- Suppose you and your teammates were paid \$1M by a big engineering company for your work. Based on your performance and the performances of your teammates, how would you distribute the money? Explain.
- What is your contribution to the group project that you are most proud of?
- Where did you struggle most in your group project?
- Review your original design proposal. How did your final product differ from what you proposed? Why?
- For future group projects, what can you do to better your team’s work?

Your individual reflection will NOT be shared with any other students in the class, so please be honest with your feedback. Additionally, your reflection has no impact on any other portion of the project.

Project Requirements

Student projects must satisfy the following three criteria:

1. The project must solve an engineering problem.
2. The project must be substantial enough to require the equivalent of three homework assignment’s worth of effort to complete.
 - Substance of the project will be judged by the instructor during the Project Proposal Meeting.
3. The project must include at least one of the advanced techniques covered in class.
 - Examples: solving systems of differential equations, optimizing a function, solving a boundary value problem.

Software Restrictions

Your code may use NumPy, SciPy, and/or Matplotlib, as well as any standard, built-in Python libraries (e.g., Random, Argparse or IO).

Within SciPy, you may only use Linear Algebra-related functions and special functions (e.g., if you want to evaluate a Bessel function) or SciPy’s built-in routines for double integrals. You may not use SciPy’s existing methods for 1D integration, optimization, or ODE solving.

If you have a question of whether a particular SciPy function may be used, please speak with the instructor.

Exception: While testing your code, you may use existing software or SciPy methods to compare your results. However, the final version of the code must be standalone.

Deadlines

Deadlines for each part of the project are summarized in the table below. All relevant submissions must be on Canvas by 11:59pm of the day indicated (unless otherwise stated).

Project Proposal	10%	February 13
Proposal Meeting and Resubmission	5%	February 27
Project Check-In Report	5%	March 27
Project Code	35%	Final Exam Period
Project Presentation	40%	Final Exam Period
Individual Reflection	5%	Final Exam Period